

include Nagios, Zabbix, Cacti, openNMS, Icinga, Op5, Munin, Network Management Information System (NMIS), NetXMS, etc.

Let us delve deeper into these tools to get an idea of their features and technicalities.

Nagios

Nagios, now known as Nagios Core, is a free, open source and powerful network monitoring tool which facilitates monitoring systems, networks and infrastructure, and ensures that all sorts of critical systems, applications and services are always up and running efficiently. Nagios Core offers monitoring and alerting services for servers, switches, applications and all sorts of network services. If any problem arises in the network, Nagios Core alerts the administrators about it and alerts them again when the issue gets resolved.

Nagios Core is regarded as the heart of the application, which comprises the core network monitoring engine and the basic Web based UI. On the top of Nagios Core, administrators can implement plugins to facilitate administrators with additional monitoring capabilities like services, applications, data visualisations, graphs and even MySQL database support.

There are various versions of Nagios:

- a. *Nagios XI* facilitates easy monitoring of mission-critical infrastructure like applications, services, operating systems, network protocols, system metrics and network infrastructure.
- b. *Nagios Log Server* simplifies the process of log data searching, as it automates the process of alerts when any potential threat is identified and quickly logs the data. The Nagios log server enables administrators to search for all sorts of network logs at one location with high availability and fault tolerance features.
- c. *Nagios Network Analyser* provides in-depth lookup of all network traffic sources and security threats, enabling systems admins to gather all the information to monitor the health of the network.
- d. *Nagios Fusion* provides network administrators with an easy and in-depth comprehensive view of multiple Nagios Core or Nagios XI servers.

Version 4.1.1 is the latest release of Nagios available for free download under GPLv2.

Listed below are the main features of Nagios Core.

- Monitors all sorts of network services like SMTP, HTTP, HTTPS, NNTP, SNMP, SSH, FTP, etc.
- Monitors all host resources like processor load, disk usage and all operating systems like Windows, Linux and their event logs.
- Remote network monitoring via Nagios Remote Plugin Executor.
- Proper data visualisation via graphs using plugins.
- Can define event handlers to run during service or host events for proactive problem resolution.

Nagios agents: These are listed below.

1. *NRPE:* Nagios Remote Plugin Executor (NRPE) allows remote system monitoring of various resources like disk usage, system load and number of users logged in.
2. *NRDP:* Nagios Remote Data Processor (NRDP) has a flexible data transport mechanism and processor, and uses standard ports and protocols (HTTP and XML).
3. *NSClient++:* This is used to monitor various services of Windows machines like memory usage, CPU load, disk usage, running processes, etc.
4. *NCPA:* Nagios Cross Platform Agent (NCPA) supports installation on Windows, MAC OS X and Linux for monitoring CPU usage, disk usage, processes, services and network usage.

Zabbix

Zabbix is regarded as an enterprise oriented open source monitoring tool for networks and all sorts of application software. It works with a centralised Linux based Zabbix server.

Zabbix is designed to do all sorts of monitoring and tracking with regard to network services, servers and various network hardware. It makes use of MySQL, PostgreSQL, SQLite, Oracle or IBM DB2 to store the data. It offers data gathering and monitoring options for servers and even supports the monitoring of virtual machines.

Architecture: Zabbix architecture is composed of three different servers/components—the Web server, the database server and the Zabbix server

In addition, using the whole Zabbix architecture in large environments allows us to have two other actors, i.e., Zabbix agents and Zabbix proxies, which also play a crucial role in efficient overall network monitoring.

Zabbix server acquires data from Zabbix proxies, which in turn acquire data from the Zabbix agents connected to them. And with all the data stored on a database server, the whole system will be monitored via a Web based UI.

The latest version of Zabbix is 3.0.3 which was released in May 2016.

The unique features of the Zabbix network monitoring system are listed below:

- Zabbix has a centralised Web interface for monitoring all servers, services and other network hardware.
- Zabbix systems are easy to integrate with other systems because of the API available in varied programming languages.
- Zabbix enables systems administrators to monitor the network via SNMP, IPMI, JMX, ODBC, SSH, HTTP, HTTPS, TCP/UDP, etc.
- Other features include: Web monitoring, secure user authentication, flexible email notifications, audit log and agent-less monitoring.
- The Zabbix monitoring system offers a wide range of customisation options for items, graphs and data visualisation.

Cacti

Cacti is regarded as a complete open source Web based